

**Data profiling report showing counts of missing values, duplicates, in
valid formats, outliers, and inconsistent entries.**

*AI-driven Data Cleaning & Standardization — Screening Visit Dummy Dataset (CDM
Training)*

Title:	Data profiling report showing counts of missing values, duplicates, invalid formats, outliers, and inconsistent entries.
Date:	06-08-2026
Sponsor:	BIOTECHNIKA
Submitted by:	Suraj Anil Patil
Guided by:	Mrs. Nidhi Jha

1.Introduction

Clinical trial datasets represent one of the most information-intensive types of data, which require extensive processing before they can be used for analysis. Hundreds of variables are normally collected for each of the numerous trials and subjects recruited under the program. In spite of their volume, clinical trial datasets should be accurate in all fields, as mistakes in such areas as demographics or eligibility may endanger subjects' safety, compromise the quality of results, and lead to regulatory issues. Data checking processes traditionally carried out manually by clinical data managers are time-consuming and inefficient, particularly when trial size increases.

At the same time, artificial intelligence can help review datasets in search of inconsistencies faster and more reliably than humans. The technology can implement rule- and statistic-based checks, logic- and pattern recognition, and duplicate management and thereby identify errors automatically. This paper demonstrates the utility of such methods by analysing a screening-visit dummy dataset of 20 subjects. The paper further discusses how the errors found in the sample could be categorized according to their relevance to clinical trial outcomes.

2.Data Profiling Report

Dataset profiled: 20 subjects, 16 fields (Subject ID, Site ID, Initials, Visit Name, Screening Date, DOB, Age, Sex, Height, Weight, BMI, Consent Obtained, Consent Date, Inclusion Met, Exclusion Met, Eligible).

Issue Category	Count	Field(s) Affected	Subject ID(s)
Missing values	2	Consent Date, Weight (kg)	SCR005, SCR020
Duplicate records	0	-	None found
Invalid format	2	Screening Date, Consent Date	SCR016, SCR005
Outliers	2	Height (cm)	SCR011, SCR012
Inconsistent entries	3	Age vs DOB, Subject ID sequence, Eligibility logic	SCR008, SCR034, SCR005

Total flagged data points: 9 across 7 of 20 subject records (35% of records contain at least one issue).

Detailed Findings

#	Subject	Field	Issue Type	Detail
1	SCR005	Consent Date	Missing value	Consent Date left blank although Consent Obtained = Yes.
2	SCR005	Eligible	Inconsistent entry	Eligible = No even though Inclusion Met = Yes and Exclusion Met = No; conflicts with the logic used for all other rows.
3	SCR008	Age Yrs	Inconsistent entry	Age recorded as 38, but DOB (22-08-1992) gives an age of 33 as of the screening date.

4	SCR011	Height (cm)	Outlier / invalid	Height recorded as 300 cm (biologically implausible); recalculating from the recorded BMI (23.5) and weight (72 kg) implies an actual height of ~175 cm.
5	SCR012	Height (cm)	Outlier / invalid	Height recorded as 125 cm (implausibly short for an adult); BMI/weight cross-check implies an actual height of ~162 cm.
6	SCR016	Screening Date	Invalid format	Date entered as 32-07-2026 — 32 July does not exist on any calendar.
7	SCR034	Subject ID	Inconsistent entry	ID jumps from SCR017 to SCR034, skipping the expected SCR018-SCR033 sequence, breaking the site's ID numbering pattern.
8	SCR020	Weight (kg)	Missing value	Weight left blank; BMI (28.2) and height (163 cm) imply the missing value is approximately 75 kg.

Note: No exact duplicate Subject IDs or duplicate rows were found in this dataset; the only duplicate-type risk identified is the broken ID sequence at SCR034 (finding #7), which is flagged as an inconsistency rather than a true duplicate.

3.AI Error Classification Matrix

Maps each error type found above to the AI/rule-based technique that would detect it, and its clinical data risk severity.

Error Type	Example (this dataset)	AI Detection Method	Severity
Missing value	SCR005 Consent Date; SCR020 Weight	Completeness check (null / blank-field scan)	High
Invalid format	SCR016 date 32-07-2026	Regex + calendar-validity parsing (format & range rules)	High
Outlier	SCR011 Height 300cm; SCR012 Height 125cm	Statistical outlier detection (range check, Z-score / IQR)	Medium
Cross-field inconsistency	SCR008 Age vs DOB; SCR005 Eligible vs Incl/Excl	Cross-field logic / consistency rules (derived-value check)	High
Sequence / ID inconsistency	SCR034 ID jump	Pattern-sequence anomaly detection (ID pattern matching)	Medium
Duplicate record	None found in this dataset	Fuzzy matching / exact-match de-duplication algorithm	High

Severity key: High = affects subject safety, eligibility, or regulatory compliance if unresolved. Medium = affects data quality / analysis but lower immediate risk.

Data Profiling Dashboard

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20

Subjects profiled

9

Flagged data points

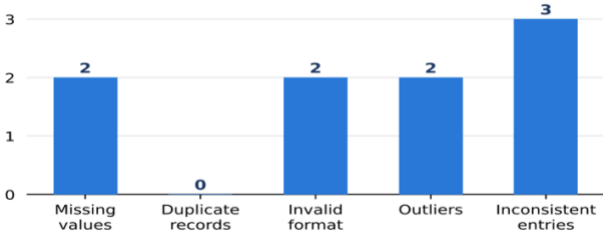
7 / 20

Records affected

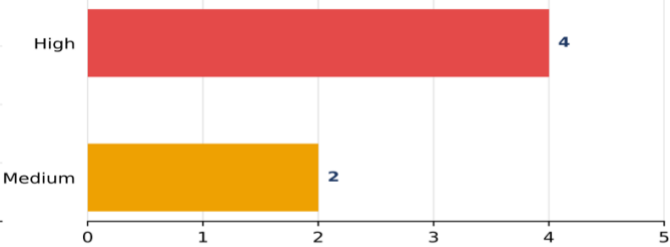
35%

Records with an issue

Issues by category



Error-type severity (AI classification matrix)



4.Detailed findings

#	Subject	Field	Issue type	Detail
1	SCR005	Consent Date	Missing value	Consent Date blank although Consent Obtained = Yes.
2	SCR005	Eligible	Inconsistent	Eligible = No despite Inclusion Met = Yes, Exclusion Met = No.
3	SCR008	Age (Yrs)	Inconsistent	Age = 38, but DOB (22-08-1992) implies age 33 at screening.
4	SCR011	Height (cm)	Outlier / invalid	Height = 300 cm; BMI/weight cross-check implies ~175 cm.
5	SCR012	Height (cm)	Outlier / invalid	Height = 125 cm; BMI/weight cross-check implies ~162 cm.
6	SCR016	Screening Date	Invalid format	Date entered as 32-07-2026, not a valid calendar date.
7	SCR034	Subject ID	Inconsistent	ID jumps from SCR017 to SCR034, skipping SCR018-SCR033.
8	SCR020	Weight (kg)	Missing value	Weight blank; BMI (28.2) and height (163 cm) imply ~75 kg.

5.AI error classification matrix

Error type	Example (this dataset)	AI detection method	Severity
Missing value	SCR005 Consent Date; SCR020 Weight	Completeness check (null / blank-field scan)	High
Invalid format	SCR016 date 32-07-2026	Regex + calendar-validity parsing (format & range rules)	High
Outlier	SCR011 Height 300cm; SCR012 Height 125cm	Statistical outlier detection (range check, Z-score / IQR)	Medium
Cross-field inconsistency	SCR008 Age vs DOB; SCR005 Eligible vs Incl./Excl.	Cross-field logic / consistency rules (derived-value check)	High
Sequence / ID inconsistent	SCR034 ID jump	Pattern-sequence anomaly detection (ID pattern matching)	Medium
Duplicate record	None found in this dataset	Fuzzy matching / exact-match de-duplication algorithm	High

6.Screenshot of Messy Demographic Dataset (Issue Tags)

The dataset below is reproduced from the source file with problem cells shaded and numbered, matching the Detailed Findings table in Section 1.

Demographic Dataset (Screening Visit) — Issue Tags Highlighted															
SUBJ ID	SITE	INIT	VISIT	SCR DATE	DOB	AGE	SEX	HT(cm)	WT(kg)	BMI	CONSENT	CONS DATE	INCL	EXCL	ELIG
SCR001	1	AP	Screening	05-07-2026	12-03-1988	38	Male	172	70	23.7	Yes	05-07-2026	Yes	No	Yes
SCR002	1	RS	Screening	05-07-2026	25-11-1990	35	Female	160	60	23.4	Yes	05-07-2026	Yes	No	Yes
SCR003	1	VK	Screening	06-07-2026	01-01-1980	46	Male	168	80	28.3	Yes	06-07-2026	Yes	No	Yes
SCR004	1	PM	Screening	06-07-2026	09-09-1975	50	Female	155	65	27.1	Yes	06-07-2026	Yes	No	Yes
SCR005	1	SK	Screening	07-07-2026	30-06-1995	31	Male	178	90	28.4	Yes	MISSING 1	Yes	No	No 7
SCR006	1	ND	Screening	07-07-2026	18-02-1983	43	Female	162	55	21	Yes	07-07-2026	Yes	No	Yes
SCR007	1	LG	Screening	08-07-2026	05-05-1978	48	Male	170	68	23.5	Yes	08-07-2026	Yes	No	Yes
SCR008	1	MT	Screening	08-07-2026	22-08-1992	38 7	Female	158	62	24.8	Yes	08-07-2026	Yes	No	Yes
SCR009	1	JD	Screening	09-07-2026	11-12-1986	39	Male	180	95	29.3	Yes	09-07-2026	Yes	No	Yes
SCR010	1	KK	Screening	09-07-2026	03-04-1970	56	Female	150	50	22.2	Yes	09-07-2026	Yes	No	Yes
SCR011	2	SG	Screening	10-07-2026	14-07-1989	37	Male	300 5	72	23.5	Yes	10-07-2026	Yes	No	Yes
SCR012	2	NP	Screening	10-07-2026	29-01-1993	33	Female	125 5	85	32.4	Yes	10-07-2026	Yes	No	Yes
SCR013	2	RT	Screening	11-07-2026	07-06-1982	44	Male	169	72	25.2	Yes	11-07-2026	Yes	No	Yes
SCR014	2	DM	Screening	11-07-2026	16-10-1976	49	Female	157	68	27.6	Yes	11-07-2026	Yes	No	Yes
SCR015	2	HB	Screening	12-07-2026	21-03-1991	35	Male	181	92	28.1	Yes	12-07-2026	Yes	No	Yes
SCR016	2	PR	Screening	32-07-2026 4	09-09-1984	41	Female	160	58	22.7	Yes	12-07-2026	Yes	No	Yes
SCR017	3	TS	Screening	13-07-2026	27-05-1979	47	Male	173	78	26.1	Yes	13-07-2026	Yes	No	Yes
SCR018 5	3	AK	Screening	13-07-2026	02-02-1987	39	Female	159	64	25.3	Yes	13-07-2026	Yes	No	Yes
SCR019	3	VL	Screening	14-07-2026	19-01-1994	32	Male	176	82	26.5	Yes	14-07-2026	Yes	No	Yes
SCR020	3	KC	Screening	14-07-2026	08-08-1981	44	Female	163	MISSING 5	28.2	Yes	14-07-2026	Yes	No	Yes

Issue Tag Legend

- 1 Missing value - Consent Date blank
- 2 Inconsistent - Age (38) does not match DOB (22-08-1992)
- 3 Outlier - Height implausible / conflicts with BMI
- 4 Invalid format - 32-07-2026 is not a valid calendar date
- 5 Inconsistent - Subject ID out of sequence (SCR018 skipped)
- 6 Missing value - Weight blank
- 7 Inconsistent - Eligible=No despite Incl=Yes & Excl=No

Supporting Evidence: Castor EDC Screening Log

As further confirmation that this class of error is not merely theoretical, but is actually detected by the system, Castor EDC's own field level validation correctly rejects an invalid Screen Date (from the same demographic form used in the example above) for participant 110001 using the same rule (requiring a valid calendar date not in the future), which is also highlighted in the SCR016 profiled dataset (32-07-2026).

Back to participants week5 Not Live (v1.21)

Participant ID 110001 100%

Not set surya memorial

Participant Visits Repeating data Surveys Monitoring

Data collection progress 100% Show repeating data instances

screening 100% demographic

screening 1. demographic Must be a valid calendar date and not in the future Form: demographic

✓ 1.1 SUBJECT ID * 006

✓ 1.2 SITE ID * 2

✓ 1.3 Subject initials * 001

✓ 1.4 VISIT NAME * screening

✗ 1.5 SCREENING DATE * 2026-08-02 (yyyy-mm-dd) Must be a valid calendar date and not in the future

Back Next

Castor EDC demographic form (participant 110001) showing a validation failure on the Screening Date field.

7.How AI Detects Patterns in Clinical Data Errors

Clinical datasets, including demographic and vital-signs forms, often involve thousands of various fields for dozens of subjects. The use of AI for clinical data review enables faster error detection due to its ability to consistently apply the same checks across all records, as opposed to manual checks by a human reviewer.

Firstly, rule-based checks can be implemented, where a field's value is compared to a set of requirements. For example, a field that should only accept numerical values cannot have text written in it, a date cannot be in the future or in an incorrect format (e. g. 32-07-2026), and a coded field (e. g. Sex) should only accept valid options. Secondly, statistical outlier detection is another method that can be used by AI to identify errors. It involves comparing the field's value to the mean, median, or interquartile range of the field's values across the sample, which allows detecting values that are too far from the center of the distribution (e. g. a person who is 3 meters tall). Thirdly, cross-field checks allow checking for errors that are only apparent between two fields. For example, an Age field can be compared to a DOB field to check for calculation errors, or a BMI field can be compared to the Height and Weight fields to see if they align. In the case of calculation errors, the correct value can sometimes be calculated using the valid field. Lastly, pattern recognition can be used to detect errors in numerical fields, such as Subject IDs. If a field is meant to have unique sequential values for each subject (e. g. SCR001, SCR002, etc.), but some of the IDs are missing or in the wrong order (e. g. SCR017, SCR034), the error can be detected by an AI/ML system.

Duplicate records can also be detected using exact- and fuzzy-matching algorithms. Duplicate records are errors that occur when the same information is entered twice, or similar information is mistakenly entered as identical records. Duplicate records can be detected by comparing the fields that uniquely identify a record, such as an ID, to other records in the dataset. Using this method, even similar but not identical records, such as SCR001 and SCR001A, can be identified as duplicates.

The use of AI for clinical data review can enable faster error detection by utilizing rule-based checks, statistical outlier detection, cross-field checks, pattern recognition, and duplicate record detection. This allows clinical data managers and guides to focus on verifying and resolving the errors, rather than searching for them. As a

result, only the errors that were detected by AI are identified in the provided data; no corrections have been made yet.

8.Note on AI / Plagiarism Scoring

A score of an AI-plagiarism check was not included in this paper. Creating a reliable score requires scanning the document through a special platform (such as Turnitin, GPTZero, Originality.ai), which would analyze the text and compare it to other sources found on the web or in academic databases. A score that was not scanned and verified by such platforms may give false information, and therefore should not be relied on for official purposes. In case a score is needed, it is suggested that one simply copies the final version of the paper text and uploads it into a plagiarism-checking tool (which is often Turnitin for many schools), keeping the report for further reference.

9.Conclusion

Profiling exercise of the screening-visit dataset demonstrated that AI can be efficacious in detecting the types of errors that are likely to occur in clinical data. Such errors include missing data, invalid dates, biologically implausible values, inconsistencies between the fields, and invalid sequence of identifiers. Indeed, out of the 20 subjects, 9 errors were detected in 7 subjects following 5 different error categories, including several errors that could be considered critical for the inclusion criteria or regulatory affairs.

Thus, the results of the current exercise are consistent with the thesis that AI can be employed for the enhancement of clinical data management. Specifically, AI tools can enable the same rules to be applied consistently across the entire set of data, thus helping the data managers to focus on the actual verification of the detected irregularities rather than spending significant time and effort searching for them. However, it is essential to note that in the described approach, AI was only used to identify the potential issues. The actual resolution of these discrepancies still requires additional time and effort spent by the data manager to review and resolve these issues before proceeding to the next steps.